

EMU660 Project

Financial Power vs Sporting Success: A 20-Year Analysis of the Turkish Super League

1. Project Overview and Scope

This project aims to analyze the relationship between financial strength and on-field performance of football clubs in the Turkish Super League over a long-term period. Football has increasingly become a data-driven industry where financial resources play a crucial role in determining competitive success. However, the extent to which financial power translates into sporting success remains an open question.

The primary objective of this study is to investigate whether higher squad values and financial indicators lead to better league performance. In addition to a single-season analysis, this project extends its scope to a longitudinal perspective by examining how clubs' financial structures and performances have evolved over the past 20 years (from the 2004–2005 season to the 2024–2025 season).

More specifically, the project will address the following questions:

- Is there a strong relationship between squad market value and league performance?
- Which clubs outperform or underperform relative to their financial capacity?

By combining financial and performance datasets, this project aims to provide both descriptive and analytical insights into the efficiency and competitiveness of clubs in the Turkish football ecosystem.

2. Data

2.1 Data Source

The datasets used in this project are collected from two primary sources:

- Transfermarkt/Startseite: Used to obtain financial and squad-related information such as total squad value, average player value, average age, squad size, and number of foreign players.
- Transfermarkt/Tabelle: Used to collect official league performance statistics such as points, wins, draws, losses, goals scored, goals conceded, and league rankings.

The data has been collected through web scraping techniques and manually verified for consistency. The Python script used can be downloaded by clicking [here](#).

The script is created thanks to Gemini 3.1 GenAI tool. You can refer to prompts used for creation of the script [here](#).

2.2 General Information About Data

The project utilizes two main datasets covering a 20-season period from 2004–2005 to 2024–2025.

- Dataset 1: Financial and Squad Data (superlig_kadro_verileri)

This dataset contains team-level financial and structural information for each season:

Takim (Team name)

Sezon (Season)

Kadro_Sayisi (Squad size)

Ortalama_Yas (Average age)

Yabanci_Oyuncu_Sayisi (Number of foreign players)

Kaleci_Sayisi (Number of goalkeepers)

Kaleci_Toplam_Deger_Euro (Total squad market value of goalkeepers)

Kaleci_Ortalama_Deger_Euro (Average squad market value of goalkeepers)

Defans_Sayisi (Number of defenders)

Defans_Toplam_Deger_Euro (Total squad market value of defenders)

Defans_Ortalama_Deger_Euro (Average squad market value of defenders)

Ortasaha_Sayisi (Number of midfielders)

Ortasaha_Toplam_Deger_Euro (Total squad market value of midfielders)

Ortasaha_Ortalama_Deger_Euro (Average squad market value of midfielders)

Forvet_Sayisi (Number of forwards)

Forvet_Toplam_Deger_Euro (Total squad market value of forwards)

Forvet_Ortalama_Deger_Euro (Average squad market value of forwards)

Ortalama_Piyasa_Degeri_Euro (Average player market value)

Toplam_Piyasa_Degeri_Euro (Total squad market value)

- Dataset 2: League Performance Data (superlig_puan_durumu_verileri)

This dataset contains performance-related statistics for each team-season:

Takim (Team name)

Sezon (Season)

Galibiyet (Wins)

Beraberlik (Draws)

Maglubiyet (Losses)

Atilan_Gol (Goals scored)

Yenilen_Gol (Goals conceded)

Averaj (Goal difference)

Puan (Total points)

2.3 Reason of Choice

The choice of this dataset is motivated by several factors:

- **Relevance to Türkiye:** The dataset focuses on the Turkish Super League, aligning with the course recommendation to use locally relevant data.
- **Richness and size:** The dataset spans 20 seasons and includes multiple variables, making it suitable for in-depth analysis.
- **Real-world applicability:** The relationship between financial investment and performance is a widely discussed topic in sports economics.
- **Availability of multiple dimensions:** The dataset allows for combining financial, demographic, and performance variables.
- **Longitudinal analysis opportunity:** The availability of long-term data enables trend analysis and structural comparisons across years.

This combination makes the dataset highly suitable for both exploratory and advanced analytical techniques.

2.4 Preprocessing

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
Rows: 387 Columns: 19
-- Column specification -----
Delimiter: ","
chr  (2): Takim, Sezon
dbl (17): Kadro_Sayisi, Ortalama_Yas, Yabanci_Oyuncu_Sayisi, Kaleci_Sayisi, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

character(0)
```

When we examined the data from the “Startseite” and “Tabelle” pages, we encountered a fundamental problem: the team names on the two pages do not match. For example, the team name listed as “A. Sebatspor” on the Startseite page appears as “Akçaabat Sebatspor” on the Tabelle page. To resolve this issue, we extracted the unique team names from both pages and mapped them to each other on a one-to-one basis. As part of this mapping process, we aligned the team names in the data retrieved from the two pages.

Datasets are stored as .csv files and imported into R. You can download the current version of the dataset in R format [here](#).

3. Analysis

In this project, multiple analyses will be conducted to investigate the relationship between financial strength and sporting performance of clubs in the Turkish Super League.

The planned analyses are as follows:

- **Correlation Analysis:** The relationship between total squad market value and total points will be examined to understand whether financially stronger teams perform better.
- **Efficiency Analysis:** Clubs will be evaluated based on how efficiently they convert financial resources into points.
- **Distribution Analysis:** The distribution of squad values and performance indicators will be analyzed to understand overall league structure.
- **Longitudinal Analysis (20-Year Club Evolution):** The squad value and performance evolution of clubs over the past 20 years will be analyzed. This includes examining trends in squad value and performance consistency, as well as identifying clubs that improved or declined over time.

3.1 Correlation Analysis

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats   1.0.1      v stringr   1.6.0
v ggplot2   4.0.2      v tibble    3.3.1
v lubridate 1.9.5      v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
Rows: 387 Columns: 19
-- Column specification -----
Delimiter: ","
chr  (2): Takim, Sezon
dbl (17): Kadro_Sayisi, Ortalama_Yas, Yabanci_Oyuncu_Sayisi, Kaleci_Sayisi, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 387 Columns: 9
-- Column specification -----
```

```

Delimiter: ","
chr (2): Takim, Sezon
dbl (7): Galibiyet, Beraberlik, Maglubiyet, Atilan_Gol, Yenilen_Gol, Averaj,...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```
[1] 387
```

```

# A tibble: 0 x 9
# i 9 variables: Takim <chr>, Sezon <chr>, Galibiyet <dbl>, Beraberlik <dbl>,
#   Maglubiyet <dbl>, Atilan_Gol <dbl>, Yenilen_Gol <dbl>, Averaj <dbl>,
#   Puan <dbl>

```

```

# A tibble: 0 x 19
# i 19 variables: Takim <chr>, Sezon <chr>, Kadro_Sayisi <dbl>,
#   Ortalama_Yas <dbl>, Yabanci_Oyuncu_Sayisi <dbl>, Kaleci_Sayisi <dbl>,
#   Kaleci_Toplam_Deger_Euro <dbl>, Kaleci_Ortalama_Deger_Euro <dbl>,
#   Defans_Sayisi <dbl>, Defans_Toplam_Deger_Euro <dbl>,
#   Defans_Ortalama_Deger_Euro <dbl>, Ortasaha_Sayisi <dbl>,
#   Ortasaha_Toplam_Deger_Euro <dbl>, Ortasaha_Ortalama_Deger_Euro <dbl>,
#   Forvet_Sayisi <dbl>, Forvet_Toplam_Deger_Euro <dbl>, ...

```

```

# A tibble: 57 x 2
  Takim          Super_Lig_Sezon_Sayisi
  <chr>          <int>
1 Beşiktaş      21
2 Fenerbahçe    21
3 Galatasaray   21
4 Trabzonspor   21
5 Kayserispor   20
6 Sivasspor     19
7 Konyaspor     18
8 Antalyaspor   17
9 Gençlerbirliği 16
10 Kasımpaşa    16
# i 47 more rows

```

```

# A tibble: 1 x 4
  Analiz          Pearson Spearman Gozlem_Sayisi
  <chr>          <dbl>    <dbl>        <int>
1 Tum takim-sezon gozlemleri  0.781    0.713          387

```

```

# A tibble: 1 x 4
  Analiz          Pearson Spearman Gozlem_Sayisi
  <chr>          <dbl>    <dbl>        <int>
1 En az 5 sezon oynayan takımlar  0.783    0.722          322

```

```

# A tibble: 21 x 4
  Sezon          Pearson Spearman Takim_Sayisi
  <chr>          <dbl>    <dbl>        <int>

```

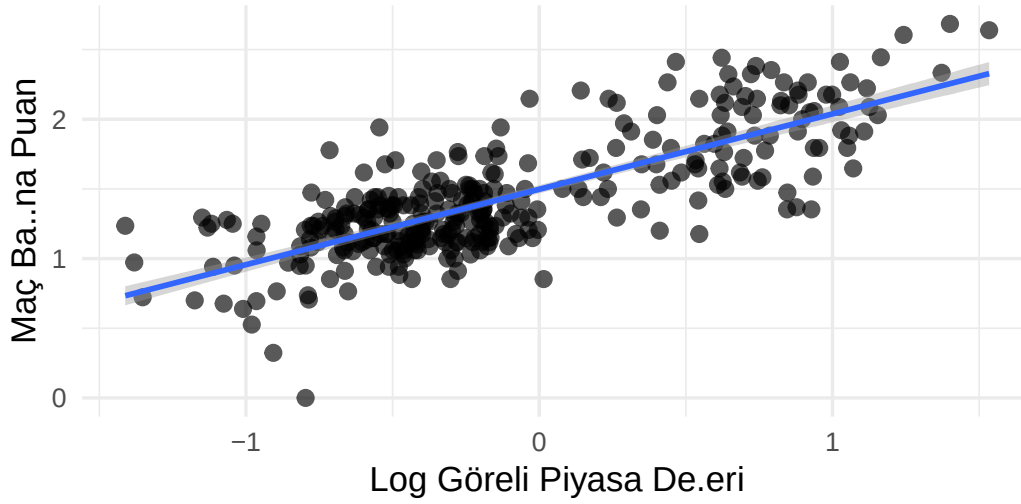
1	2004-2005	0.871	0.789	18
2	2005-2006	0.758	0.453	18
3	2006-2007	0.771	0.702	18
4	2007-2008	0.792	0.618	18
5	2008-2009	0.758	0.731	18
6	2009-2010	0.715	0.769	18
7	2010-2011	0.755	0.799	18
8	2011-2012	0.822	0.848	18
9	2012-2013	0.734	0.839	18
10	2013-2014	0.768	0.635	18

i 11 more rows

```
`geom_smooth()` using formula = 'y ~ x'
```

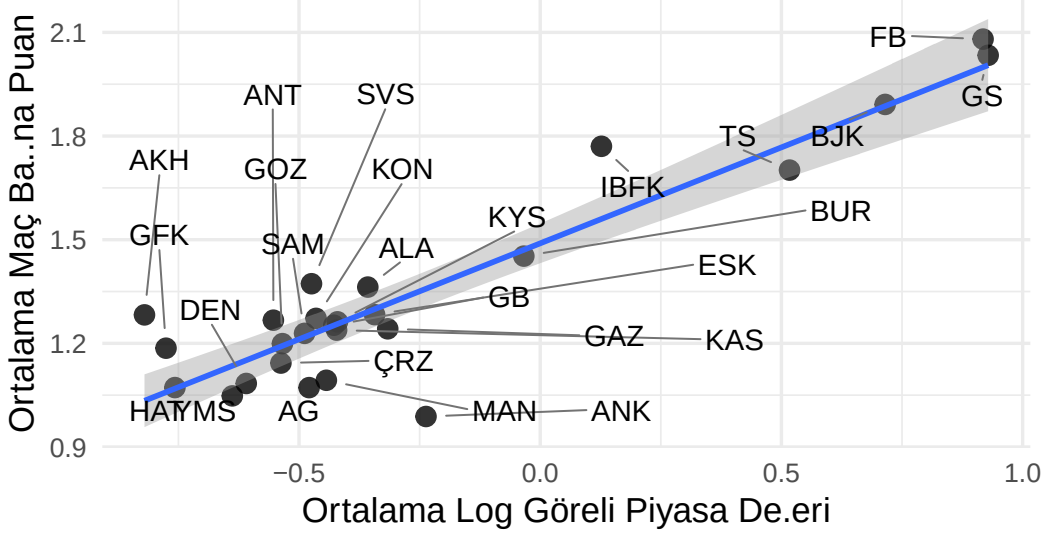
Kadro Piyasa De.eri ile Puan Performans. Aras.nda

X: log(sezon ortalamas.na göre gö.reli piyasa de.eri), Y: ma. ba
Grafikte yaln.zca en az 5 sezon Süper Lig'de yer alan tak.mlar ı



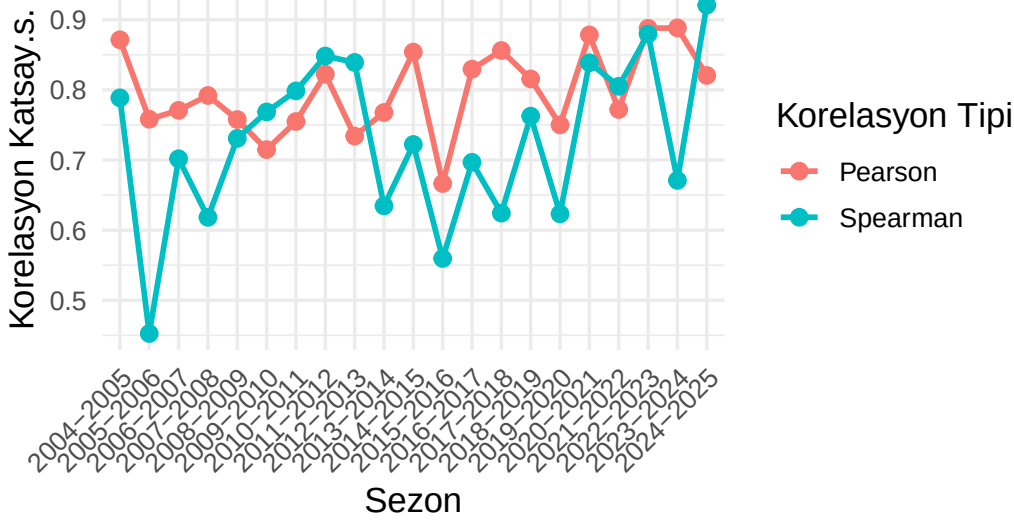
```
`geom_smooth()` using formula = 'y ~ x'
```

Takım Bazında Ortalama Kadro Değeri ve Ortalama Yalnızca en az 5 sezon Süper Lig'de yer alan takımlar gösterildi



Sezon Bazlı Korelasyon Değerleri

Her sezon için tüm takımlar kullanılmıştır.



Due to economic inflation over time and various factors related to the football industry, the purchasing power of 1 million euros in football in 2004 is not the same as that of 1 million euros in 2024. If we had simply taken squad market value figures into account, all teams from 2004 would have appeared “super efficient,” while today’s teams would have appeared “very inefficient.” For this reason, we looked at **relative squad market value** instead of squad market value. This value is calculated by dividing each team’s squad market value by the league’s average squad market value for that season. Additionally, since squad market value figures contain extreme variations, we examined the **logarithmic transformation** of relative squad market value figures to mitigate this extremity.

The number of teams in the league varied across some seasons. While the number of teams per season was mostly 18, in the season when relegation was suspended due to COVID-19 and subsequent seasons aligned with that format, the league featured 19, 20, or 21 teams. For this reason, we factored in **points per game** rather than total points.

A team that has only spent a few seasons in the Turkish Super League but managed to accumulate a decent number of points by securing 3–5 wins along the way (e.g., Pendikspor or Balıkesirspor) could

have topped the “most efficient team” list due to mathematical anomalies. For this reason, to filter out “yoyo clubs,” we included the criterion of “teams that have competed in the Turkish Super League for at least 5 seasons.”

3.1.1 Interpreting relationship between “Log Görelî Piyasa Değeri” and “Maç Başına Puan”

Call:

```
lm(formula = Mac_Basina_Puan ~ Log_Goreli_Piyasa_Degeri, data = veri_grafik)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.06636	-0.17133	-0.02028	0.16621	0.73866

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.49734	0.01492	100.36	<2e-16 ***
Log_Goreli_Piyasa_Degeri	0.54112	0.02401	22.53	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2631 on 320 degrees of freedom

Multiple R-squared: 0.6134, Adjusted R-squared: 0.6122

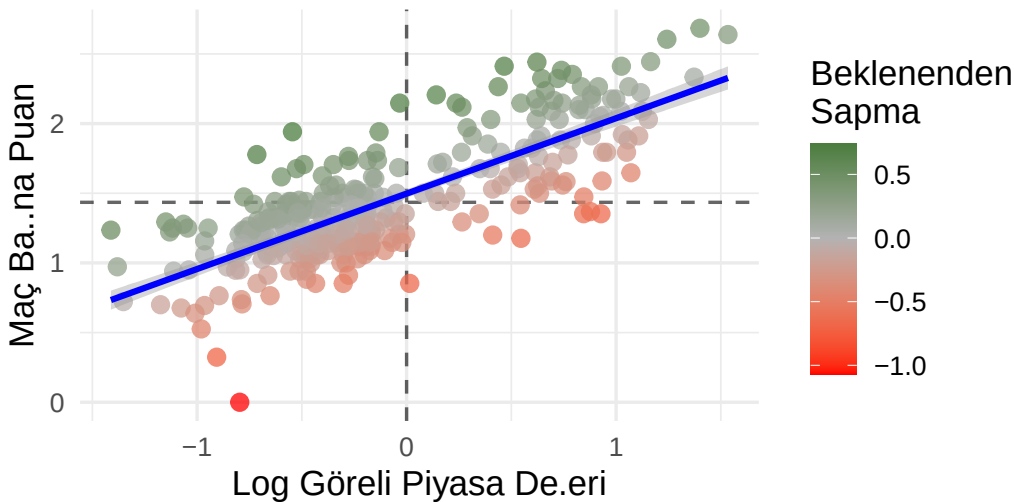
F-statistic: 507.8 on 1 and 320 DF, p-value: < 2.2e-16

```
`geom_smooth()` using formula = 'y ~ x'
```

Kadro Piyasa Değeri ile Puan Performans. Arasında

Renkler regresyon çizgisine göre beklenenden iyi/kötü performan

Ye.il: beklenenden iyi, k.rm.z.: beklenenden kötü.



The graph shows a positive correlation between teams’ relative squad value during the season and their points-per-game performance. This suggests that financially stronger squads generally produce higher athletic performance. However, the significant scatter of data points around the regression line indicates that squad value alone is not sufficient to explain success. A value of 0 on the log-scale relative market value axis indicates that a team has a market value equal to the league average for the relevant season. The fact that the majority of the data points do not cluster around 0 suggests a

significant divergence in squad values within the Turkish Super League. Points above the regression line represent team-seasons that demonstrate higher-than-expected point performance relative to squad value; points below the line represent team-seasons with lower-than-expected performance. Therefore, residual analysis can be used to measure sporting efficiency relative to financial strength.

3.1.2 Interpreting “not-yoyo clubs” (Teams that participate in Turkish Super League for at least 5 years)

Call:

```
lm(formula = Ortalama_Mac_Basina_Puan ~ Ortalama_Log_Goreli_Deger,
    data = takim_ortalama_pozisyon)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.36976	-0.06244	0.00396	0.06408	0.24845

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.48947	0.02787	53.44	< 2e-16 ***
Ortalama_Log_Goreli_Deger	0.55536	0.05008	11.09	6.28e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1248 on 24 degrees of freedom

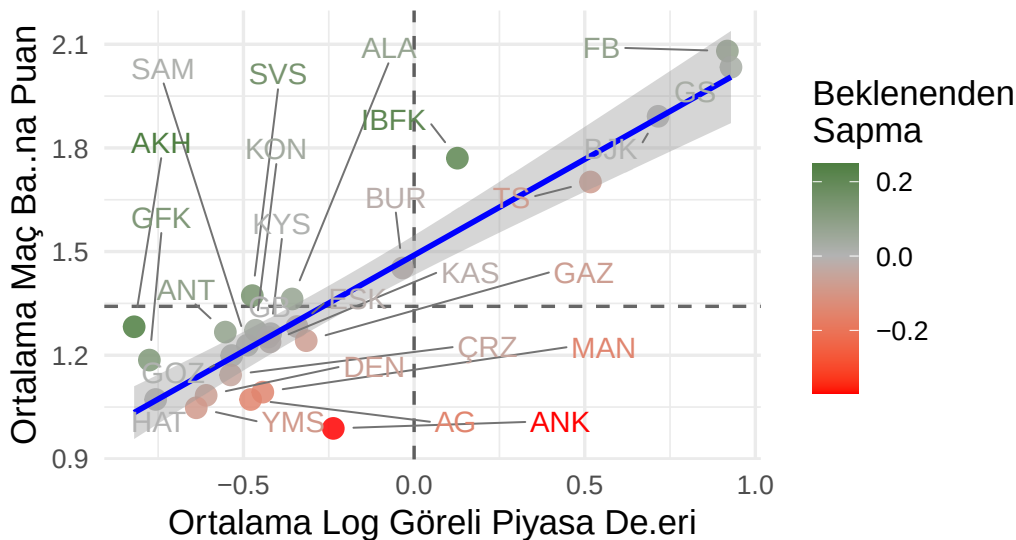
Multiple R-squared: 0.8367, Adjusted R-squared: 0.8299

F-statistic: 123 on 1 and 24 DF, p-value: 6.283e-11

`geom\_smooth()` using formula = 'y ~ x'

Takım Bazında Kadro Değeri ve Puan Performansı.

Renkler regresyon çizgisine göre beklenenden iyi/kötü uzun d



When examining average values by team, it is clear that Fenerbahçe, Galatasaray, and Beşiktaş rank among the top tier of the league in terms of both average relative market value and average points per game. These teams' high point averages are largely consistent with their high squad values.

Although Trabzonspor ranks in the top tier in terms of average squad value, it lies below the regression line. This indicates that, over the 20-year period, Trabzonspor's expected points performance fell slightly short of what would be expected given its squad value.

On the other hand, Başakşehir and Akhisar are clearly positioned above the regression line. This demonstrates that clubs have produced a significantly higher points performance relative to their average squad values and serves as a successful example of converting financial strength into points.

The presence of numerous teams in the left half of the graph indicates that squad values in the Turkish Super League are concentrated in favor of a few major clubs, and that many clubs are competing with market values below the league average. The relatively balanced distribution of teams on the left side of the line (both above and below) highlights significant differences in sporting efficiency among low and mid budget clubs.

3.1.3 Interpreting correlation coefficients

Pearson correlation is the most commonly used statistic to measure the degree of the relationship between linearly related variables.

For the Pearson correlation, both variables should be normally distributed, since normally distributions exhibit a bell-shaped curve. Additionally, other key assumptions include linearity and homoscedasticity. Linearity assumes a straight line relationship between each of the two variables, while homoscedasticity assumes that data is equally distributed about the regression line.

Spearman correlation is a non-parametric test that measures the degree of association between two variables. Moreover, the Spearman rank correlation test does not carry any assumptions about the distribution of the data. It is the appropriate correlation analysis when one measure the variables on a scale that is at least ordinal.

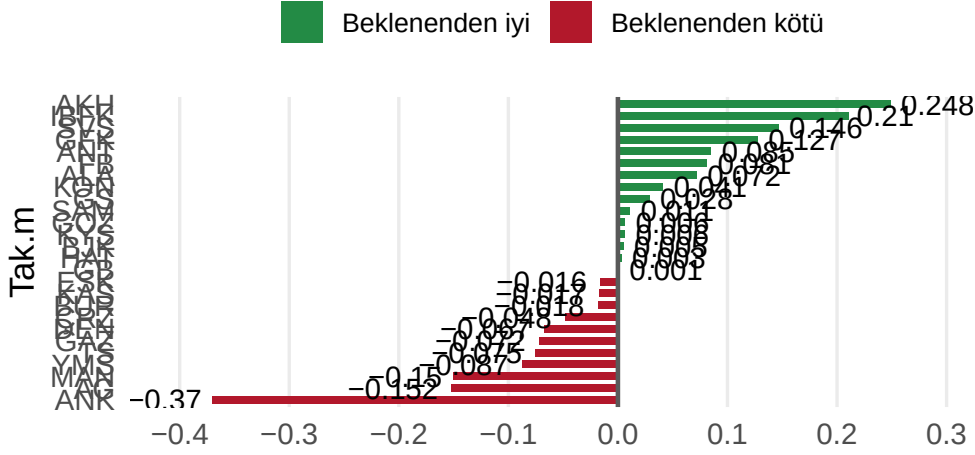
For our context, Pearson correlation answers: "In each season, is there a linear relationship between PPG and log of relative squad market value?" and Spearman correlation answers: "In each season, are ranks of points collected and squad market value similar?"

- When examining season-based correlations, it is clear that the relationship between squad market value and points performance is generally positive and strong. This suggests that teams with greater financial strength tend to accumulate more points in most seasons. However, differences between Pearson and Spearman correlations reveal that the nature of this relationship varies across seasons. In seasons where the Pearson correlation coefficient is significantly higher than the Spearman correlation coefficient, while the linear relationship between squad value and points is strong, the teams' market value rankings do not fully align with their point standings. This can be explained by some teams performing better or worse than expected relative to their squad values.
- Special case when **Pearson » Spearman** (Examples: 2005-2006, 2017-2018 and 2023-2024)
 - There is a strong linear relationship between squad value and points; however, the squad value rankings and the points rankings are not aligned to the same extent. There are several possible reasons for this:
 - * A few high-value teams may have accumulated very high points, thereby skewing the Pearson correlation upward.
 - * In the middle-to-lower ranks, squad value and point rankings may be mixed up.
 - * Two or three teams in the league may be significantly divergent in terms of both financial and sporting performance; this could boost the linear correlation while weakening the overall alignment of the rankings.

3.2 Efficiency Analysis

Kadro Değerine Göre Beklenenden Sapma

Pozitif değerler kadro değerine göre beklenenden iyi, negatif değerler beklenenden kötü



Ölçülen Sapma: Gerçek Ortalama Maç Başına Puan – Beklenen Ortalama Puan

Teams with positive residuals outperformed the expected points-per-match level implied by their average squad market value, while teams with negative residuals underperformed relative to that expectation. Therefore, the residual ranking measures not only overall sporting success, but also how efficiently clubs converted their financial and squad-value strength into points.

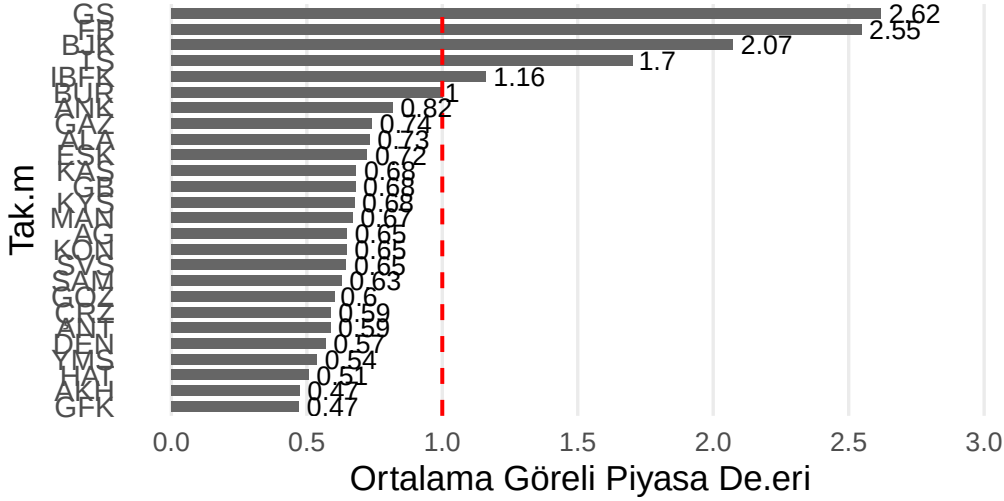
In this ranking, clubs such as **Akhisar Belediye**, **Başakşehir**, and **Sivasspor** stand out on the positive side, indicating stronger-than-expected long-term performance relative to their squad values. In contrast, **Ankaraspor**, **Ankaragücü**, **Manisaspor**, and **Yeni Malatyaspor** appear on the negative side, suggesting that they collected fewer points than expected given their average squad values.

Overall, the results show that long-term Süper Lig performance cannot be explained by squad value alone. The residual metric therefore provides a useful measure for comparing clubs' sporting efficiency relative to their financial strength.

3.3 Distribution Analysis

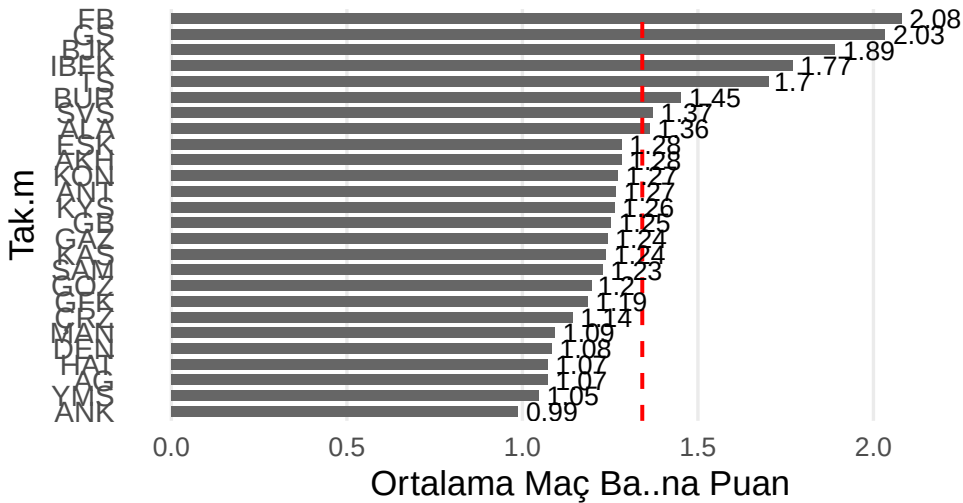
Tak.m Baz.nda Ortalama Görelî Piyasa D

Yaln.zca en az 5 sezon Süper Lig'de yer alan tak.mlar gösterilmi.t
Kesikli çizgi, sezon ortalamas.na e.it de.eri göstermektedir.



Tak.m Baz.nda Ortalama Maç Ba..na Pua

Yaln.zca en az 5 sezon Süper Lig'de yer alan tak.mlar gösterilmi.t
Kesikli çizgi, bu tak.m grubunun ortalama maç ba..na puan de.eri

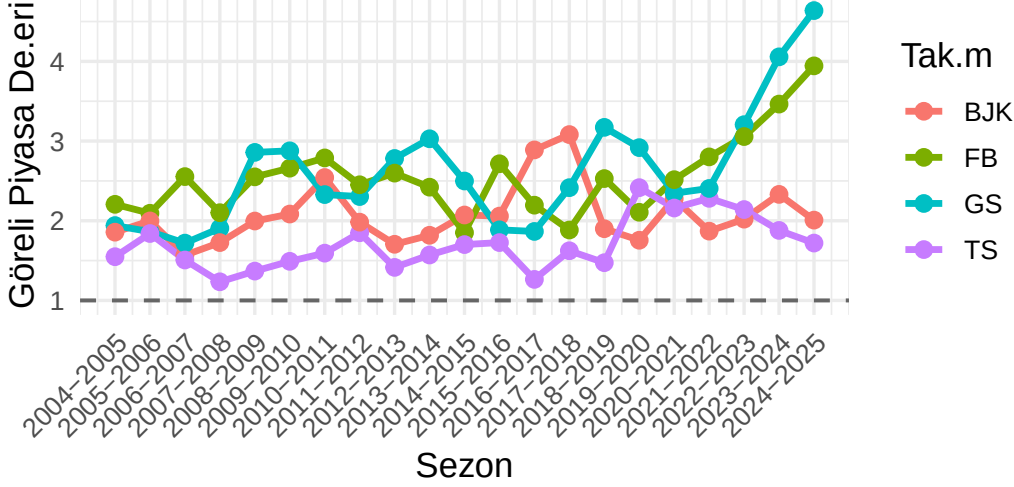


3.4 Longitudinal Analysis

```
# A tibble: 4 x 2
  Takim      Sezon_Sayisi
  <chr>      <int>
1 Beşiktaş      21
2 Fenerbahçe    21
3 Galatasaray   21
4 Trabzonspor   21
```

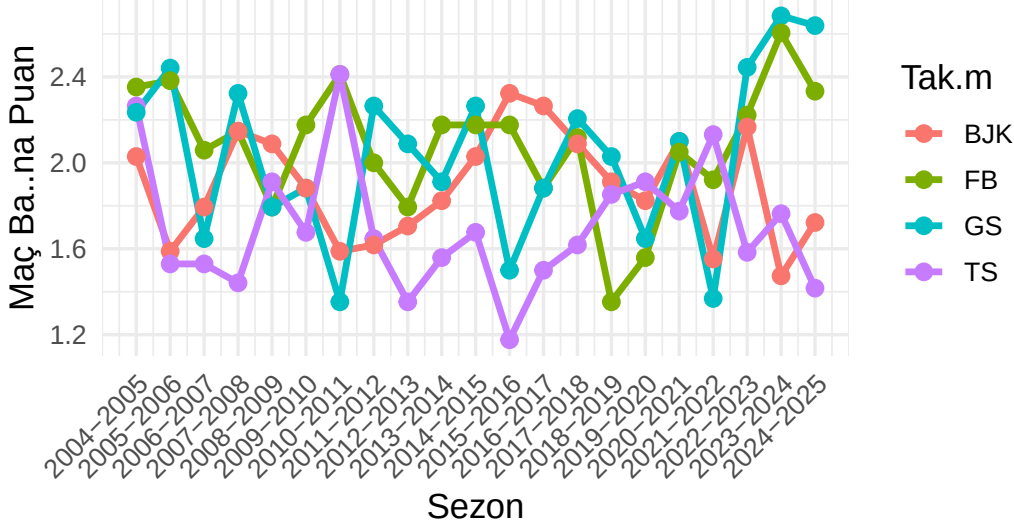
20 Y.İI.k Göreli Kadro De.eri Evrimi

Yaln.zca veri setindeki tüm sezonlarda yer alan tak.mlar gösteri
Kesikli çizgi, ilgili sezonun lig ortalama kadro de.erini gösterir.



20 Y.İI.k Maç Ba..na Puan Evrimi

Yaln.zca veri setindeki tüm sezonlarda yer alan tak.mlar göste



A tibble: 84 x 7

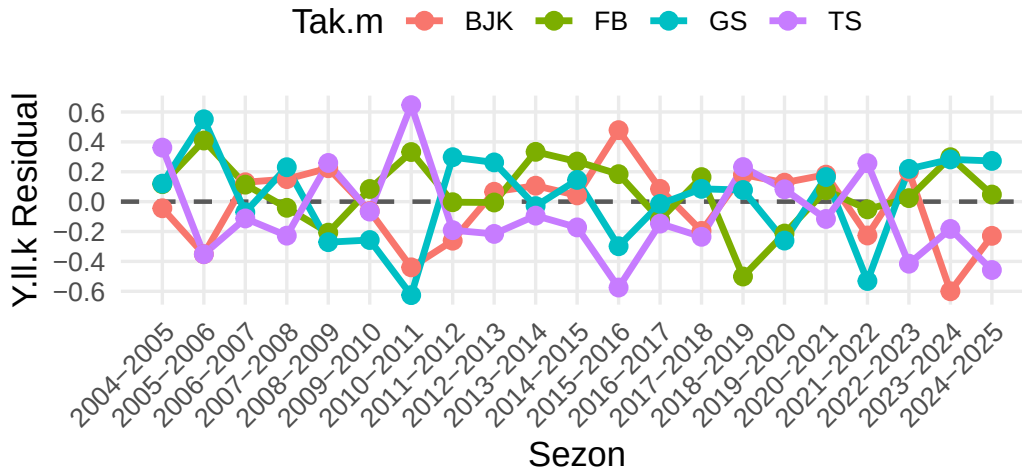
	Sezon	Takim	Takim_Kisa	Log_Goreli_Piyasa_Degeri	Mac_Basina_Puan
	<chr>	<chr>	<chr>	<dbl>	<dbl>
1	2004-2005	Beşiktaş	BJK	0.619	2.03
2	2005-2006	Beşiktaş	BJK	0.692	1.59
3	2006-2007	Beşiktaş	BJK	0.449	1.79
4	2007-2008	Beşiktaş	BJK	0.546	2.15
5	2008-2009	Beşiktaş	BJK	0.691	2.09
6	2009-2010	Beşiktaş	BJK	0.734	1.88
7	2010-2011	Beşiktaş	BJK	0.933	1.59
8	2011-2012	Beşiktaş	BJK	0.684	1.62
9	2012-2013	Beşiktaş	BJK	0.533	1.71
10	2013-2014	Beşiktaş	BJK	0.597	1.82

i 74 more rows

```
# i 2 more variables: Beklenen_Mac_Basina_Puan <dbl>, Residual <dbl>
```

Dört Büyük Tak.m.n Y.İl.k Verimlilik De.İ.i

Residual = Gerçek maç ba..na puan – kadro de.erine göre beklen
Pozitif de.erler beklenenden iyi, negatif de.erler beklenenden kötü



3.5 Key Takeaways